Medicare Claims Analysis: Cost and Utilization by Chronic Condition Profile

An analysis of synthetic Medicare inpatient claims using BigQuery, Power BI, and Python

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From 2008 to 2010, members with all three tracked conditions (diabetes, kidney disease, and heart failure) accounted for 38% of plan spend, cost about $27K per member, and were readmitted within 30 days 15.6% of the time, compared with 4.0% among members with none of these conditions. They account for a disproportionate share of spending and readmissions, which makes it a useful starting point for outreach and case management efforts.

What the data shows

Spend is concentrated in the three-condition group. Of $639M in total plan spend, this group accounts for 38%. Its average spend per member is about $27K, less than half of that for members with no tracked conditions. Kidney disease appears in the next two highest groups (kidney and heart failure, and diabetes and kidney disease). The three-condition group also averages about 6.4 days per stay.

Readmissions show a similar pattern. The overall 30-day readmission rate is 10.0%, and the three-condition group made up 36% of eligible stays but 57% of readmissions. I examine readmissions in more detail in “30-Day Readmissions by Chronic Condition Profile”.

Age does not explain the difference in cost per claim. Claim volume peaks for members 65 to 84, but average cost per claim stays about the same across age groups.

The spend trend needs a caveat. Annual plan spend falls from about $255M in 2008 and $248M in 2009 to $138M in 2010. Claim volume drops sharply in the fourth quarter of 2010 by both admission date and discharge date. The drop appears to reflect incomplete synthetic data, so I would not interpret it as a genuine decline in spending.

How I built it

I loaded the CMS DE-SynPUF beneficiary and inpatient claims files into BigQuery and built two tables: a member table with a health profile based on the diabetes, kidney disease, and heart failure flags, and a claims table with admission and discharge dates and payment amounts. I wrote SQL for spend and utilization by health profile, age group, and length of stay, flagged high-utilization members using the 95th percentile of claim counts, and calculated year-over-year spend. The Power BI dashboard is built on those tables.

I then rebuilt the main results in Python (pandas), connected to BigQuery. The totals by gender and by health profile, and the average length of stay, matched the SQL results.

Data checks

226 claims (0.3%) have late-2007 admission dates. I kept them because they were processed and discharged in 2008.

A small group of members has a placeholder birth date of 1909-01-01. I put them in their own age group so they would not inflate the 85 and older group, and left them off the dashboard's age chart.

Some claims have negative plan paid amounts. I looked at five members' claims for reversals (a negative claim matching a positive claim's amount and date) and found none, so I left the claims in. I treat them as a quirk of the synthetic data. There are no null values in the payment amounts.

Limits

The dataset is synthetic, so the results demonstrate the analytical method rather than real-world cost or utilization patterns. Health profiles come from the 2008 enrollment file and are applied to claims from 2007 to 2010, and claims from members missing from that file are dropped by the inner join. The 2010 data is incomplete. The negative claim amounts are unexplained.